Big Data Analytics - Module-wise Numbered Questions

Module 1: Introduction to Big Data and Hadoop

1. What is the basic difference between traditional RDBMS and Hadoop? [5]

2. Mention four characteristics of Big Data and explain in detail. [5]

3. What are the 3 V?s of Big Data? Give two Big Data case studies indicating respective V?s with justification. [5]

4. Explain how Big Data problems are handled by Hadoop system. [5]

5. Explain CAP. How is CAP different from ACID property in databases? [5]

6. Describe the components of Hadoop ecosystem with the help of a diagram. [10]

7. Why is HDFS more suited for applications having large datasets and not when there are small files? Elaborate. [5]

metadata overhead : many small files
poor storage utilisation :
low processing efficiency :

8. Secondary Name Node is a backup of Name Node. Is this statement True or False? Justify your answer. [5]

Module 2: Hadoop HDFS and MapReduce

1. Explain the function of Map Tasks in the MapReduce framework with the help of an example. [5]

2. Write a MapReduce pseudo code for Word Count problem. Illustrate with example showing all steps. [10]

3. Write a MapReduce pseudo code to multiply two matrices. Apply MapReduce working to perform matrix multiplication. [10]

4. Explain the MapReduce execution pipeline with a suitable example. [10]

5. Explain selection and projection relational algebraic operations using MapReduce. [10]

6. Explain natural join and grouping and aggregation relational algebraic operation using MapReduce. [10]

7. Draw and explain the architecture of a Data-Stream Management System. [10]

Module 3: NoSQL

1. List and explain the core business drivers behind the NoSQL movement. [5]

volume
veracity
Velocity
agility

2. What is a Key-Value Store? What are the benefits of using a Key-Value Store? [10]

3. What is Graph Store? Give an example where it can be effectively used to solve a particular business problem. [10]

4. Describe the four ways by which Big Data problems are handled by NoSQL. [10]

vast data (volume)
real time data (velocity)
scalability (horizontally scalable)
flexible schema
BASE properties

5. Write a short note on variations of NoSQL architectural patterns. [10]

6. List the architectural patterns in NoSQL databases. Discuss Key-Value and Document-Oriented patterns with examples. [10]

7. Demonstrate how business problems have been successfully solved faster, cheaper, and more effectively considering NoSQL Google?s MapReduce or BigTable case study. [5/10]

Module 4: Mining Data Streams

1. Explain the concept of Bloom Filter with an example. [5/10]

2. Create a Bloom Filter with the following parameters (bit array size, hash functions, etc.) and perform insertions and queries. [10]

3. Suppose the stream S = {4, 2, 5, 9, 1, 6, 3, 7}. Using hash function h(x) = 3x + 7 mod 32, show how the Flajolet-Martin algorithm estimates distinct elements. [10]

4. List down six constraints that must be satisfied for representing a stream by buckets using DGIM algorithm. [5]

5. Explain DGIM algorithm for counting ones in a stream with an example. [10]

6. Explain issues and challenges in data stream query processing. [5]

7. Explain selection and projection relational algebraic operation using MapReduce. [10]

Module 5: Real-Time Big Data Models

1. Determine communities for the given social network graph using Girvan-Newman algorithm. [10]

2. Write an algorithm for the Clique Percolation Method and discover communities in a given graph (k=3). [10]

3. How is collaborative filtering used in recommendation systems? [10]

4. Define collaborative filtering. Using an example of an e-commerce site like Amazon or Flipkart, describe how it provides recommendations. [10]

5. What is a recommendation system? How is classification algorithm used in recommendation systems? [10]

6. Justify the use of a Content-Based Recommendation System with a specific case study. [10]

7. How recommendation is done based on properties of product? Explain with example. [10]

Module 6: Data Analytics with R

1. Create a Data Frame in R for given project data and display the output. [10]

2. Update the Data Frame to include new projects and demonstrate the output. [10]

3. Write a script to create a dataset named data1 in R containing the following text: 2, 3, 4, 5, 6.7, 7, 8.1, 9. [10]

4. Explain functions provided by R to combine different sets of data. [10]

5. List and discuss various types of data structures in R. [10]

6. Write the script to sort values in a vector (e.g., (23,45,10,34,89,20,67,99)) in ascending and descending order. [10]

7. Name and explain operators used to form data subsets in R. [10]

8. Create a subset of data frame using brackets and subset() function with given conditions. [10]

9. Explain basic features of R, RGUI, RStudio, handling data in workspace, using functions, importing/exporting data. [10]

10. Describe applications of Data Visualization. [10]

Uncategorized Questions

1. Name three ways resources can be shared between computer systems. Name the architecture used in Big Data solutions and describe it in detail. [10]

2. Describe applications of Data Visualization (sometimes cross-module between R and visualization topics). [10]

3. Describe the cost of exact counts and query answering in DGIM or Datar-Gionis algorithms (if applicable in variations). [10]